蔡秉樺 Ping-Hua Tsai

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Summary

M.S. graduate in Aeronautics and Astronautics from National Cheng Kung University (NCKU) with hands-on experience in satellite attitude control, embedded systems, and hardware-software integration. Proficient in C++, MATLAB/Simulink, and embedded platforms, with experience in firmware development, sensor integration, system debugging, and hardware validation.

Education

M.S. in Aeronautics and Astronautics, *National Cheng Kung University* (*NCKU*) 08/2024 – 08/2026
Research Focus: Control Systems

B.S. in Aeronautics and Astronautics, *National Cheng Kung University* (*NCKU*) 09/2020 – 07/2024

Skills

Programming:
C/C++, MATLAB, Python

Embedded Systems:
Arduino, Teensy, Raspberry Pi, Sensor Integration

System Engineering:
- Hardware-Software Integration
- System Debugging
- Hardware Validation

Projects

NCKU Gemini Satellite Team, *ADCS & EPS Subsystem Development*
- Tested and verified ADCS components, including reaction wheels, sun sensors, and magnetometers, and supported attitude control validation using an air-bearing platform.
- Performed hardware-software integration and communication debugging between the onboard computer (OBC) and ADCS modules.
- Conducted battery discharge and thermal tests, evaluated solar panel performance, and debugged EPS communication and telemetry interfaces.

2024 CanSat Competition, *Gemini Space | Electrical Engineering Team Lead*
- Designed a custom PCB for the CanSat electrical system, integrating sensor interfaces and supporting electronic components.
- Developed test programs to verify sensor communication and data acquisition, while troubleshooting hardware connections and PCB interfaces.
- Coordinated with the mechanical team on component placement and system assembly, and performed basic functional testing.

Additional Information

Availability: January 2027